

Statistics Hybrid Practice Set – S11 (L7)

Topics Covered

- Population
- Sample
- Mean
- Median
- Mode
- Range
- Population Variance
- Population Standard Deviation
- Quartiles (Q1, Q2, Q3)
- IQR
- Outlier Detection
- Frequency Table
- **Histogram (New)** ★

Instructions

- Solve on **paper first**.
 - Then solve the **same questions in Excel**.
 - Show your calculations where required.
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Q1. Identify Population and Sample

A university has **25,000 students**. It randomly selects **900 students** for a survey.

Paper Task

- Population = ?
- Sample = ?

Excel Task

Total Students	Surveyed Students
25000	900

Q2. Calculate the Mean

Value
35
40
45
50
55

Paper Task

- Calculate the Mean.

Excel Task

- Verify your answer.
-

Q3. Find the Median

Value
56
40
48
44
52

Paper Task

- Find the Median.

Excel Task

- Verify your answer.
-

Q4. Complete Statistical Analysis

Value
14
18
22
26
30

Paper Task

Calculate:

- Mean
- Range
- Population Variance
- Population Standard Deviation

Excel Task

Verify all answers.

Q5. Quartiles & IQR

Value
9
11
13
15
17
19
21
23
25

Paper Task

Find:

- Q1
- Q2
- Q3
- IQR

Excel Task

Verify your answers.

Q6. Outlier Detection

Value
32
34
36
38
40
42
44
46
95

Paper Task

Find:

- Q1
- Q3
- IQR
- Lower Fence
- Upper Fence

Is **95** an outlier?

Excel Task

Verify your answer.

★ Q7. Frequency Table

The following exam scores were obtained by **24 students**.

Scores
42
45
47
48
50
53
55
57
59
61
63
65
67
69
71
73
75
77
79
82
84

Scores
86
88
92

Create a frequency table.

Class Interval	Frequency
40–49	?
50–59	?
60–69	?
70–79	?
80–89	?
90–99	?

Excel Task

Create the same table using **COUNTIFS()**.

★ Q8. Histogram (New)

Using the **frequency table from Q7**:

1. Create a **Histogram** in Excel.
2. Which class interval has the **highest bar**?
3. Which class interval has the **lowest bar**?
4. Does the data appear:
 - Mostly evenly distributed
 - Left-skewed
 - Right-skewed

Write one sentence explaining your answer.

Note: For now, create the histogram using the frequency table you've built. We'll learn Excel's automatic Histogram chart in a later set.

Q9. Real Data Analyst Scenario

A school principal wants to quickly understand **how student marks are distributed across score ranges** and present the information visually in a meeting.

Which visualization is the **best** choice?

- A. Pie Chart
- B. Histogram
- C. Line Chart
- D. Scatter Plot

Explain your answer in one sentence.

Q10. Mixed Business Case

A company records the following delivery times (minutes):

18, 19, 20, 20, 21, 22, 45

Answer:

1. Is there likely an outlier?
2. Which method would you use to confirm it?
3. Which measure (Mean or Median) would better represent the typical delivery time?
4. Explain your answer in one sentence.